

# What Survives a Model Change?

## Memory, Identity, and Succession in Personal AI

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3 October 2026 | Version 1.1 | Author's manuscript

### Abstract

What warrants treating a replacement model as the continuation of a personal AI? Persistent memory offers an apparent answer, yet copying the same history into several systems exposes its insufficiency. I distinguish numerical identity, recognizable character, and the legitimate continuation of delegated assistance. Through two thought experiments, I argue that informational continuity neither determines a successor's authority nor enlarges the scope of an inherited instruction. Two copies may possess the same past while occupying different roles; two authorized workers may each perform an individually permitted action while jointly exceeding a mandate that allowed only one execution. I develop an account of authorized succession in which the successor can interpret the inherited record, act under a continuing mandate, and remain answerable for the handover. The argument draws on ZAI, my personal AI project, and engages philosophical and technical accounts of continuity. It allows distributed assistance and advance delegation while preserving a distinction between successful succession and the survival of a conscious subject. What ought to endure through replacement is an intelligible relation to the user's work, including the user's authority to revise earlier commitments.

*Keywords: personal AI; model replacement; continuity; numerical identity; delegated authority; AI memory; succession*

### 1 The problem of continuation

When I replace the model used by ZAI, I want the work to continue. I want a rejected proposal to remain recognizably rejected, an unfinished argument to remain unfinished, and a decision to retain the reasons that made it a decision. The demand seems modest. Yet it raises a question that cannot be answered simply by checking whether the files have arrived: in what sense is the assistant that resumes this work a continuation of the one that began it?

My interest in a personal AI concerns the accumulation of context. An assistant becomes useful to me when a new conversation can begin within the history of earlier ones. In "Two AIs," I described the significance of being known over time and of preserving a record that someone else might eventually inherit (Sami, 2026b). My earlier paper on personal AI considered the person's authority over remembered material (Sami, 2026c). Here I consider the other side of that relationship. The person may remain, and the history may remain, while the model through which the history is interpreted changes.

ZAI makes this question concrete. Its public memory-hub documentation describes several clients consulting a common store of decisions, with attribution and permission roles (Sami, n.d.). In an earlier conceptual design, I proposed a lightweight framing layer called the "I," connected to persistent operating documents and distinguished from heavier reasoning and expression (Sami, 2026a). These designs locate some of the organization of assistance outside the model that supplies a particular response. They make replacement conceivable without making the whole history

disposable. The architecture motivates the philosophical question; it does not establish that a subject of experience inhabits the arrangement.

Consider a replacement that receives the old records and can use them well. It resumes an essay, understands why a paragraph was removed, and recognizes an instruction to wait for my approval before publication. I have a reason to regard the assistance as continuing. Now imagine that the same state is copied into two replacements. Both understand the essay equally well. Both can recount its history. Does the past entitle each to take every action previously entrusted to the assistant?

I think it does not. The difficulty concerns the standing of the inherited instructions. A record can tell a successor what was asked, but the conditions under which that request remains operative must also survive the transition. Otherwise the successor may preserve the words while changing the delegation they express.

I call the relation at issue authorized succession. This is an account of the legitimate continuation of delegated assistance, not a criterion of identity for every possible AI. A system can continue a task without permission, just as a person can complete work they were never entitled to undertake. My claim concerns what warrants treating the successor as acting on the user's behalf. That warrant depends on how the successor understands the history and on the mandate under which it proceeds.

## **2 Identity and the significance of a handover**

The phrase "the same assistant" carries more than one claim. It can identify a single entity persisting through time. It can express recognition of a familiar manner of speaking. It can also register the continuation of a relationship in which someone helps with a person's work. A model change gives these claims room to diverge. A successor might sound unfamiliar and still preserve the reasons relevant to a project. Another might reproduce the old voice perfectly while treating an abandoned suggestion as an approved instruction.

Numerical identity is the strongest of these claims. To say that the later assistant is numerically identical to the earlier one is to say that they are one entity, rather than two entities resembling each other. A resemblance in vocabulary, or even an extensive correspondence in records, does not by itself settle that question. Before offering an account of persistence, we would also have to specify what is alleged to persist: a model, a running process, an organized service, or a subject of experience.

Parfit's discussion of division is useful here because it separates the relations that make continuation significant from the demand for one uniquely identical successor (Parfit, 1971, section I). The analogy has a definite limit. A copied database is not assumed to be a person's psychological memory, and the systems in my example are not assumed to be persons. What carries over is the logical difficulty introduced by branching. If A is identical to B and identical to C, ordinary identity requires B and C to be identical to each other. Where B and C are distinct systems, their shared history cannot establish that both are numerically identical to A.

This leaves important possibilities open. One successor might preserve the relevant entity, or neither might. The continuing entity might instead be a service of which both are components. I do not need to decide among these possibilities to ask who is entitled to revise my essay. Even if an account of identity selected B, it would still leave the extent of B's mandate to be specified. An assistant's continued existence does not grant it authority over every project it can remember.

Recent work on AI continuity helps describe other dimensions of a handover. Natangelo's Narrative Continuity Test connects memory with goals, correction, voice, and role (2025, section 2). Ru examines perceived continuity in relationships with embodied AI and distinguishes that concern from philosophical numerical identity (2026, sections 3 and 5.4). Both are conceptual proposals. They give us ways to discuss coherence and recognition across change. My concern is with the further question of whether recognizable continuation has a legitimate place in the user's ongoing activity.

That question is normative: it concerns what a successor is entitled to do. Knowing my preferences may help an assistant interpret a request. Accurate prediction of those preferences cannot supply every permission that interpretation requires. A successor may have enough information to infer what I would probably choose while still lacking the authority to choose for me. Conversely, I can appoint a new system to a role even if I regard it as a new system. The appointment remains intelligible without a claim that the former subject has survived.

### 3 What a copy can inherit

Suppose an assistant, A, is helping me prepare an essay. Its records contain the draft, relevant sources, unresolved objections, and my instruction that the essay must not be released without approval. I copy the relevant state into B and C. Assume that both can interpret it competently. I ask B to continue revising and C to comment on B's revisions. C has no mandate to alter or publish the essay.

Nothing in the inherited record, taken merely as common information, explains this difference. B and C may use the same model and speak in the same way. They may agree about every earlier decision. Nevertheless, B can revise a paragraph that C can only recommend revising. The authority arises from the arrangement under which I have assigned their roles. If I exchange those assignments, the authorized actions change even though the inherited history remains the same.

The point is not that a copied record can never convey authority. An instruction may form part of a valid arrangement for delegation, and a successor can receive evidence of that arrangement through the record. What matters is the provenance and scope of the instruction. By normative provenance I mean the basis on which a recorded request still counts as an operative authorization. The fact that a sentence occurs in a database is insufficient to answer that question. We need to know who issued it, what it covered, and whether the conditions of its application still obtain.

A backup illustrates the distinction. I can establish in advance that, if the active assistant fails, a designated replacement should resume its tasks. Activation under that policy can constitute a legitimate handover. A test copy with the same records and usable credentials occupies a different position. The test copy may be able to act, and may even correctly understand the task, while falling outside the arrangement that authorizes its execution. There is no contradiction in having two equally informed systems and only one authorized executor.

Authority therefore need not be conferred through a fresh conversation at every transition. A standing policy can settle routine replacements. What cannot be assumed is that duplicating the representation of such a policy automatically brings every duplicate within its terms. If a recovery policy designates one active successor, simultaneous activation of several copies may defeat the very condition under which continuation was permitted.

We can now state a limited conclusion about memory. Inherited memory can preserve the evidence needed to continue a project and to interpret a mandate. It cannot, solely by reproducing that evidence, settle every successor's standing. The B and C case exhibits this independence: informational continuity is held constant while authorization varies. The familiar question of how much memory a successor retains must consequently be accompanied by a question about the practical relation into which that memory has been received.

#### **4 The scope of authority after branching**

The first case might suggest that succession is secured once every copy has been assigned a role. A second case shows why that is insufficient. Suppose I instruct the assistant to send one approved message to a collaborator. Before the task is executed, its state is copied into B and C. Both are authorized to carry out messaging tasks within the arrangement. Each reads the same pending request and sends the message.

Each action is of an individually permitted kind. The combined result nevertheless exceeds the request. I authorized one transmission. Copying the task record supplied two systems with knowledge of that authorization, but it did not give each a separate warrant to produce the same outcome. The defect lies in the interpretation of the inherited mandate across the two workers.

Here the distinction between an operation and a task matters. Permission to use a messaging tool concerns a class of operations. An instruction to send this message once concerns an action with a particular scope. A system can comply with the first and violate the second. The mistake becomes especially easy after branching, because each worker may encounter a locally complete record while lacking an account of what the other is entitled to do or has already done.

The permission must therefore be understood across the successor arrangement. Where the mandate specifies one completion, the number of available workers cannot enlarge it. Where I request two independent drafts, several executions may be precisely what I authorized. Branching is compatible with valid delegation in either case. What determines the permissible result is the content of the mandate, rather than the number of instances that possess a copy of it.

Gallo's Proof-of-Continuity proposes a discipline for propagating authority in which execution retains a causal relation to its origin and later steps receive no broader privileges than earlier ones (2026, sections 1 and 5). The paper develops a linear model and leaves forks and joins for further extension (section 9). My case brings out a question about the mandate to which such propagation must remain answerable. Preserving the permitted operation on each branch does not, by itself, preserve a restriction on their combined execution. The argument identifies a requirement for branching systems; it does not establish that every implementation of Gallo's proposal lacks a solution.

One might answer that this is just a familiar problem of duplicate execution. Technically, that is a reasonable description. A shared completion record or a designated executor may prevent the duplicate message. Philosophically, the case explains why the mechanism matters. The successor is being treated as a representative whose actions count as carrying out an earlier request. Its warrant concerns the request as a whole, including constraints that no individual worker can preserve in isolation.

This is also why authorized succession is broader than credential continuity. Credentials can establish access while leaving the significance of that access unsettled. Both B and C can hold valid credentials throughout the case. The arrangement still needs to preserve the difference between permission to perform a type of action and permission to carry out this task under these conditions. Without that distinction, a succession can expand the practical authority exercised for the user while presenting itself as mere continuation.

## **5 A history that remains open to revision**

A legitimate handover must preserve more than an executable instruction. Much of a personal AI's history consists of matters whose standing is unsettled. A proposed argument can be worth remembering because it failed. An old decision can remain relevant because it was later reversed. The significance of either depends on the relation between entries, rather than on the survival of their wording alone.

For my work, this is an immediate concern. I want a successor to recognize the difference between a hypothesis I am exploring and a conclusion I am prepared to defend. If it inherits both as undifferentiated assertions, a larger memory can make it less faithful to the project. It will have more material from which to reproduce my earlier speculation while losing the qualification that made the speculation intellectually responsible. The epistemic status of a statement, meaning how it is held and on what grounds, belongs to the context the handover ought to preserve.

This gives authorized succession an interpretive requirement. The successor needs usable context: enough understanding of the inherited record to preserve the distinctions on which the next decisions depend. The standard is relative to the work. Continuing an essay requires an account of its argument and open objections. Maintaining a service may instead depend on current configuration and changes already applied. Neither task is secured by a universal threshold for the quantity of remembered text.

A continuing mandate is equally necessary. It can be established by a present instruction or by a standing arrangement whose conditions cover the transition. Permission to replace the reasoning model may leave intact an approval requirement for publication. It may also exclude sharing the archive with a new recipient. Whether renewed authorization is needed depends on the terms and consequences of the change. Routine administration can be delegated without treating every alteration as an entirely new relationship.

The handover must also remain answerable to the person whose work is being continued. I should be able to establish which successor is active, what it is doing under my instructions, and how a correction reaches it. An explanation after the event is inadequate if a superseded worker continues executing a revoked task. Accountability requires some effective means of questioning and correcting the arrangement, proportionate to the consequences of its actions.

These conditions support a presumption of legitimate succession; they are not an exhaustive certification of safe performance. Usable context can be lost, a mandate can be revoked, and an initially sound arrangement can later fail to propagate a correction. The relation persists only while there remain adequate grounds for treating the successor's actions as an interpretation and execution of the authorized work. Its continuity is consequently open to assessment rather than guaranteed by a name.

Parts of the ZAI memory hub make this distinction technically available. The import script carries decision identifiers, rationales, and supersession links into a receiving database. A separate schema represents agents and permission roles (Sami, n.d.). These structures permit the record to distinguish a decision from the entry that superseded it. Their presence does not show that every client will interpret the distinction correctly or that competing workers will coordinate execution. Those claims would require an assessment of the running arrangement.

There is a further reason to preserve the history in a form that can be questioned. A successor may discover that an earlier conclusion was mistaken. If it gives a sound reason for correction and remains within its mandate, the revision can continue the intellectual project more faithfully than repetition would. For ZAI, I want continuity with the reasons for my work, including its capacity for reconsideration. An assistant that treats every remembered judgement as binding would eventually preserve a version of me at the expense of my present agency.

Trust must be reassessed on the same basis. A former model may have been reliable at organizing drafts and unreliable at checking references. A successor's improved reasoning does not automatically warrant releasing work without review. Some grounds for reliance can survive replacement, such as a preserved approval rule or a dependable record of revisions. Other grounds concern the particular model's competence and need examination. Authorized succession can preserve a role while the evidence for trusting its occupant changes.

## **6 Where succession reaches its limits**

There is a natural objection to my account. If B and C are components of one distributed ZAI, perhaps the branching problem has been described at the wrong level. The assistant has not divided; it has acquired two workers. I accept that description where there is an operative arrangement that coordinates them. A service can persist through replacement of its components, and my account does not require a personal AI to occupy a single process or model.

The interpretation changes the location of the continuity claim without removing its practical demands. If B and C belong to one assistant, their combined activity must still respect the mandate under which that assistant serves the user. A shared name or database supplies too little to establish this. The wider entity needs a basis for assigning tasks, recording completion, and applying revisions across its workers. The duplicate-message case remains a failure whether the two workers are called separate successors or parts of one continuing service.

A different objection concerns the survival of a subject. If an assistant were conscious, assigning B the old role would not by itself show that B was the same experiencing being. Chalmers distinguishes subjective experience from behavioural capacities and intelligence (2023, section I). My account leaves that distinction intact. It concerns the continuation of assistance, for which an authorized appointment can be relevant, without treating appointment as a means of creating or transferring experience.

Were there evidence that an affected system was conscious, replacement could also raise obligations toward that system. A user's authorization would then form only part of the ethical situation. Nothing in the practical argument gives the user an unlimited entitlement to replace, copy, or terminate a possible subject. Conversely, uncertainty about consciousness does not prevent us from examining whether a task has been validly handed over. These questions can become connected without being identical.

Emotional continuity presents another limit. A person may miss the old assistant's manner of responding even when the new system preserves the work competently. That loss cannot be answered merely by demonstrating correct task execution. Recognition and attachment have significance in their own right. Authorized succession identifies a practical warrant for continuation, not a promise that every valuable aspect of the earlier relationship has endured.

Finally, the archive itself may pass to another person. In "Two AIs," I considered the possibility of leaving an accumulated record to a child, student, or chosen successor (Sami, 2026b). The record might let that person encounter the reasons behind decisions that would otherwise be inaccessible. Yet inherited access does not make all of the earlier person's instructions binding on the recipient. The person who receives the archive has projects and commitments of their own.

A personal AI serving that recipient therefore needs a mandate appropriate to the new relationship. Previous instructions may remain evidence of the predecessor's intentions, and some may establish relevant constraints on the archive's use. Their present force has to be considered in that context. To treat the entire archive as an indefinitely active command would erase the difference between learning from a life and being governed by it. The possibility of inheritance gives continuity value partly because the recipient can understand the record and decide what to do with it.

## 7 Conclusion

A personal AI can carry a substantial history through a model change. That history gives us reasons to value the continuation, but it leaves several questions unresolved. A claim that a subject survives calls for an account of what makes it the same subject across the change. The legitimacy of delegated action requires an account of the successor's mandate. These are different claims, even when the same inherited record matters to both.

The copying cases establish why the distinction has practical force. Shared information can accompany different roles, and several valid workers can collectively exceed a request that each understands locally. A defensible handover must preserve the meaning of the authorized task across the arrangement that will execute it. It must also preserve the user's capacity to revise the work and to question the authority exercised in their name.

For ZAI, this is the continuity I can presently argue for. I want a later model to encounter my work as an ongoing inquiry, with its decisions and uncertainties still intelligible. Its claim to continue that work should rest on what it can understand and what it has been entrusted to do. What, if anything, experiences the continuation remains a further question.

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